

GA4 Website Traffic Performance Analysis Report

A comprehensive data analytics project using Python & Power BI

Data Period: May 1, 2024 – May 31, 2024 | Data Source: Google Analytics 4
Tool Stack: Python (Pandas, Matplotlib, Seaborn) | Dashboard: Microsoft Power BI

47.2K	57.5K	248.7K	32.6K	01:32
Total Users +12.5% MoM	Sessions +9.3% MoM	Events +18.6% MoM	Engaged Sessions +11.7% MoM	Avg Eng. Time -4.8% MoM

1. Executive Summary

This report presents an end-to-end analysis of website traffic data exported from Google Analytics 4 (GA4) for the month of May 2024. The dataset contains approximately 3,182 hourly records spanning five traffic channels: Direct, Organic Search, Organic Social, Referral, and Unassigned.

The project employs Python for data wrangling and exploratory data analysis (EDA), and Microsoft Power BI for interactive business dashboard creation. Key findings indicate strong overall growth across all core metrics month-over-month, with specific opportunities identified in social channel engagement and UTM parameter governance.

2. Data Overview & Preprocessing

2.1 Dataset Description

The raw CSV export from GA4 contained embedded header rows and mixed data types requiring structured cleaning before analysis.

Column	Type	Description
channel group	Categorical	Traffic acquisition channel
DateHour	DateTime	Timestamp (YYYYMMDDHH format)
Users	Integer	Unique users in the period
Sessions	Integer	Total sessions
Engaged Sessions	Integer	Sessions with meaningful engagement
Avg. Engagement Time	Float	Mean time on site (seconds)
Engaged Sessions/User	Float	Engagement depth per user
Events per Session	Float	Average events per session
Engagement Rate	Float	Engaged / Total sessions ratio
Event Count	Integer	Total events fired

2.2 Preprocessing Steps

- Removed embedded header rows and reset the index
- Renamed all 10 columns to consistent, readable names
- Parsed DateHour from YYYYMMDDHH string format to Python datetime objects
- Cast all numeric columns using `pd.to_numeric()` with `coerce` for safety
- Engineered new features: Hour (int), Date, Non-Engaged Sessions
- Verified zero null values across all key columns after cleaning

3. Channel Performance Analysis

3.1 Traffic Volume by Channel

Direct traffic dominates in raw user volume, contributing 21.2K users (44.9% of total). Organic Search is the second-largest source at 15.7K users, while Organic Social, Referral, and Unassigned trail significantly.

Channel Group	Users	Sessions	Eng. Sessions	Eng. Rate	Share of Users
Direct	21,200	25,800	15,792	61.3%	44.9%
Organic Search	15,700	19,100	13,001	68.1%	33.3%
Organic Social	6,100	7,900	2,822	35.7%	12.9%

Channel Group	Users	Sessions	Eng. Sessions	Eng. Rate	Share of Users
Referral	3,600	3,800	1,716	45.2%	7.6%
Unassigned	600	900	166	18.4%	1.3%

3.2 Engagement Quality by Channel

Engagement rate is a critical quality metric. Organic Search outperforms all other channels with a 68.1% engagement rate, confirming that SEO-driven traffic arrives with genuine intent. Direct traffic, despite its volume, shows a 61.3% rate — solid but with room for improvement. Organic Social at 35.7% and Unassigned at 18.4% represent the largest improvement opportunities.

4. Temporal Analysis

4.1 Daily Traffic Trends

Sessions and users track closely throughout May 2024, confirming consistent sessions-per-user ratios across the month. Traffic peaked around May 15-16 and May 22-23, likely corresponding to content publication or marketing campaign activations. A notable dip around May 27-28 suggests weekend seasonality effects.

4.2 Hourly Traffic Patterns

Analysis of the hour-by-channel heatmap reveals consistent traffic concentration between 10:00 and 14:00 across all channels. Direct traffic shows the broadest distribution, indicating bookmarked/returning visitors across all hours. Organic Social traffic shows a distinct evening secondary peak (19:00-21:00), aligning with social platform usage patterns.

Time Window	Peak Traffic Channels	Observation
06:00-09:00	Direct, Organic Search	Morning commute — branded queries
10:00-14:00	All channels (peak)	Core business hours — highest volume
14:00-17:00	Direct, Organic Search	Afternoon — sustained but declining
19:00-21:00	Organic Social	Evening social browsing peak
22:00-05:00	Low across all channels	Overnight low — minimal traffic

5. Engagement Analysis

5.1 Engagement Rate Trends

The dual-axis time series of engagement rate vs. sessions reveals an inverse relationship on high-traffic days: when sessions spike, engagement rate tends to dip slightly. This is consistent with campaign-driven traffic influxes where a portion of new visitors do not engage meaningfully. The overall average engagement rate sits at approximately 56% for the period.

5.2 Engaged vs. Non-Engaged Sessions

Direct channel contributes the largest absolute volume of both engaged and non-engaged sessions. However, Organic Social has a disproportionately high non-engaged session share relative to its total — meaning social visitors are frequently bouncing without meaningful interaction. This points to a disconnect between social content and landing page experience.

Channel	Engaged	Non-Engaged	Engagement Rate
Organic Search	13,001	6,099	68.1%
Direct	15,792	10,008	61.3%
Referral	1,716	2,084	45.2%
Organic Social	2,822	5,078	35.7%
Unassigned	166	734	18.4%

6. Python EDA Visualizations

The Jupyter Notebook (website.ipynb) generates 7 publication-ready charts using Matplotlib and Seaborn:

#	Visualization	Chart Type	Key Finding
1	Sessions & Users Over Time	Line	May peak ~May 15-16
2	Total Users by Channel Group	Bar	Direct leads at 21.2K
3	Avg. Engagement Time by Channel	Bar	Referral highest avg time
4	Engagement Rate Distribution	Box Plot	Organic Search most consistent
5	Engaged vs. Non-Engaged Sessions	Stacked Bar	Social has poor engagement %
6	Traffic Heatmap (Hour x Channel)	Heatmap	Peak: 10:00-14:00

#	Visualization	Chart Type	Key Finding
7	Engagement Rate & Sessions Over Time	Dual-Line	Inverse spike relationship

7. Power BI Dashboard

The interactive Power BI dashboard translates all analytical findings into a business-ready reporting interface with real-time filter capabilities.

7.1 Dashboard Components

- **KPI Cards (5):** Users, Sessions, Events, Engaged Sessions, Avg. Engagement Time — each showing MoM % change with trend arrows
- **Channel Group vs. Users:** Horizontal bar chart with value labels for quick volume comparison
- **Channel Group vs. Engagement Rate:** Clustered vertical bar chart highlighting quality differences across channels
- **Date Hour vs. Users:** Time-series line chart revealing daily user patterns across May 2024
- **Date Hour vs. Sessions:** Companion time-series for session volume trending
- **Filter Panel:** Date Range slider, Channel Group dropdown, Device Category, and Source/Medium filters

7.2 Design Principles

- Consistent blue colour palette aligned with GA4 brand guidelines
- Trend indicators (green up-arrows, red down-arrows) for instant KPI reading
- All charts use data labels to eliminate the need for axis reading
- Responsive filter panel allowing slice-and-dice by any dimension

8. Recommendations & Next Steps

Priority	Area	Action	Expected Impact
HIGH	UTM / Tagging	Audit all campaign URLs; fix unassigned traffic	Eliminate 1.3% untracked traffic
HIGH	Organic Social	Redesign social landing pages; add clear CTAs	+15-20% engagement rate on social
HIGH	Peak Hour Campaigns	Schedule pushes at 10:00-14:00 daily	Maximise reach efficiency

Priority	Area	Action	Expected Impact
MEDIUM	Avg. Engagement Time	A/B test page layouts; reduce bounce triggers	Recover the 4.8% MoM drop
MEDIUM	Referral Partnerships	Scale top referral sources with content syndication	Grow 7.6% channel share
LOW	Direct Traffic CRO	Personalise returning visitor journeys	Lift 61.3% eng. rate toward 68%+

9. Technology Stack

Tool / Library	Version	Purpose
Python	3.10+	Core analysis language
Pandas	2.0+	Data wrangling and aggregation
NumPy	1.24+	Numerical computations
Matplotlib	3.7+	Base charting engine
Seaborn	0.12+	Statistical visualizations and styling
Jupyter Notebook	7.0+	Interactive analysis environment
Microsoft Power BI	Desktop	Interactive business dashboard
Google Analytics 4	—	Data source (GA4 export)

10. Conclusion

This project successfully demonstrates a complete analytics pipeline — from raw GA4 CSV data through Python-based EDA to an executive-level Power BI dashboard. The analysis reveals strong overall growth momentum for the website in May 2024, with a 12.5% increase in users and 18.6% growth in events month-over-month.

The most actionable insight is the quality gap between Organic Search (68.1% engagement rate) and Organic Social (35.7%) — representing an immediate opportunity to improve social landing page relevance and CTAs. Fixing the unassigned traffic tagging issue and leveraging the 10:00-14:00 traffic peak window for campaign scheduling are high-priority, low-effort wins.

Future extensions of this project could include predictive modelling of traffic trends, cohort analysis of returning users, and automated reporting via the GA4 API and Python scheduling.

